PAPER_682: Numerical UQFF Stability Analysis for Sagittarius A*

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Framework: Unified Quantum Field Framework (UQFF) v5.30
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Class: `UQFFStabilityNumericallyForSgrA` | CP4 #266
Source: `grok_share_fc21e30c24b4.txt`
Domain: Black Hole Stability / UQFF

Abstract

We perform a four-pronged numerical stability analysis of the UQFF solution for Sgr A* ($M = 4.297 \times 10^6\, M_\odot$): (1) perturbation expansion with imaginary frequency $\omega_I^{\mathrm{UQFF}} < 0$ (damped), (2) Lyapunov exponent $\lambda_{\mathrm{UQFF}} = -(\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}})e^{-U_m/k_B T_H}/\tau_{\mathrm{std}} < 0$, (3) RK4 mass evolution $M(t)$ confirming quasi-static behaviour over 10^{60} s, (4) fixed-point stability classification.

Primary UQFF Equation

$$\lambda_{\mathrm{UQFF}} = -\frac{\rho_{\mathrm{SCm}}}{\rho_{\mathrm{UA}}}\frac{e^{-U_m/k_B T_H}}{\tau_{\mathrm{std}}(M)} < 0$$

Parameters

Parameter	Value	Description
$M_{\text{Sgr A*}}$	$4.297 \times 10^6\, M_\odot$	Black hole mass
T_H	1.4×10^{-14} K	Hawking temperature
λ	< 0	Lyapunov exponent

UQFF Constants

Constant	Value	Description
ρ_{UA}	7.09×10^{-36} kg/m^3	Universal Aether density
ρ_{SCm}	7.09×10^{-37} kg/m^3	Schwarzschild condensate
f_{TRZ}	0.1	Time-reversal zone factor
κ	5×10^{-4} day^{-1}	UQFF calibration constant
$[\mathrm{SSq}]$	0.57	Superstring quenching factor
μ_J	3.38×10^{23} J·m	Magnetic string coupling
γ	5×10^{-5} / 86400 s^{-1}	Aether oscillation decay

C++ Implementation

```cpp
#include "UQFFStabilityNumericallyForSgrA.h"

UQFFStabilityNumericallyForSgrA obj;
auto result = obj.compute_primary(...); // primary equation
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```python
from CondensedPhysics2 import UQFFStabilityNumericallyForSgrACalculator

calc = UQFFStabilityNumericallyForSgrACalculator()
result = calc.compute()  # returns all UQFF quantities
print(result)
```

References

Event Horizon Telescope 2022 (Sgr A*); UQFF Session 173

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